

Paper Club · Sep 9

Visionary

11 + 3 minutes

2026 bar · follow-ups · don't overclaim

arXiv 2512.06951 · 1st Place BEHAVIOR 2025

Seat 3

2026 Bar · Still ~26% Partial

Room to grow — don't overclaim

partial q_score

~26%

binary only ~11-12%

Long-horizon household bimanual remains largely unsolved

Invest in recovery demos + closed-loop chunking benchmarks,
not only raw success rate

Challenge design: single ckpt + ablations appendix + with/without-heuristics reporting

Follow-ups · Research Agenda

Recovery data

Failure-state demos
as first-class track

Outer-loop System 2

VLM checklists /
video understanding

Unify to 1 ckpt

Multi-task vs 4 FT
with longer train

Heuristics reporting

Require with/without
in leaderboard

Corrective offline RL

Distill recovery
into VLA student

Privileged teacher

Sim RL → distill;
keep real honest

Semantic stages

Beyond timestamp
stage IDs

Dexterity residuals

+ real-robot FM
inpaint vs RTC

Motion / Control Spine · Not classical planning

Learned action-chunk FM + rolling receding-horizon + stage hierarchy + rule recovery



Closed-loop overrides (outer loop)

System 2 · Stage Voting

5-15 stages / task
vote window = 3 ($\geq 2/3$ forward)
rollback rules for aliasing

NON-MARKOVIAN context

Gripper Recovery Heuristic

force open if closed in stage
never closed in demos
paper: $\sim 2.2 \times q$ on 13×3 eps

CLOSED-LOOP OVERRIDE

Challenge design asks

Ablations appendix · single ckpt track · with/without heuristics

Don't overclaim

Winning stack $\neq$ settled science · push concrete follow-ups

Visionary close

Recovery data · VLM outer loops · real-robot RTC